

Mathematics Reference: Mensuration

Educational Series

December 29, 2025

1 2D Shapes (Area and Perimeter)

1.1 Triangles

- **General Triangle:** Area = $\frac{1}{2} \times \text{base} \times \text{height}$
- **Heron's Formula:** Area = $\sqrt{s(s-a)(s-b)(s-c)}$
where $s = \frac{a+b+c}{2}$ (semi-perimeter).
- **Equilateral Triangle:** Area = $\frac{\sqrt{3}}{4}a^2$; Height = $\frac{\sqrt{3}}{2}a$

1.2 Quadrilaterals

- **Square:** Area = a^2 ; Perimeter = $4a$; Diagonal = $a\sqrt{2}$
- **Rectangle:** Area = $l \times b$; Perimeter = $2(l + b)$; Diagonal = $\sqrt{l^2 + b^2}$
- **Parallelogram:** Area = base $\times$ height
- **Rhombus:** Area = $\frac{1}{2} \times d_1 \times d_2$; Perimeter = $4a$
- **Trapezium:** Area = $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$

1.3 Circle

- **Area:** πr^2
- **Circumference:** $2\pi r$
- **Area of Sector:** $\frac{\theta}{360} \times \pi r^2$
- **Length of Arc:** $\frac{\theta}{360} \times 2\pi r$

2 3D Shapes (Surface Area and Volume)

Note: For a cone, slant height $l = \sqrt{r^2 + h^2}$.

Shape	Lateral/Curved S.A.	Total S.A.	Volume
Cube	$4a^2$	$6a^2$	a^3
Cuboid	$2h(l + b)$	$2(lb + bh + hl)$	$l \times b \times h$
Cylinder	$2\pi rh$	$2\pi r(r + h)$	$\pi r^2 h$
Cone	πrl	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

3 Important Conversion Units

- $1 \text{ cm}^2 = 100 \text{ mm}^2$
- $1 \text{ m}^2 = 10,000 \text{ cm}^2$
- $1 \text{ hectare} = 10,000 \text{ m}^2$
- $1 \text{ litre} = 1000 \text{ cm}^3$
- $1 \text{ m}^3 = 1000 \text{ litres} = 1 \text{ kilolitre}$

4 Solved Example

Problem: Find the volume and total surface area of a cylinder with radius 7 cm and height 10 cm. (Use $\pi = 22/7$)

Solution:

1. **Volume** $= \pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 10 = 1540 \text{ cm}^3$
2. **Total S.A.** $= 2\pi r(r + h) = 2 \times \frac{22}{7} \times 7 \times (7 + 10) = 44 \times 17 = 748 \text{ cm}^2$